CRYSTALLOGRAPHY & X-RAY DIFFRACTION — GRADUATE EXAM

Total Points: 50.0 | Questions: 5 | Date: March 29, 2026

Diagram-based questions on Crystallography & X-ray Diffraction

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Question 1 10.0 points

Examine the diagram below illustrating the geometry of Bragg's law, where an incident X-ray beam strikes a crystal plane and the diffracted beam is observed. Identify the angle of incidence, the spacing between planes, and the conditions for constructive interference according to the equation $n\lambda = 2d\sin\theta$. Explain how each component is represented in the diagram.

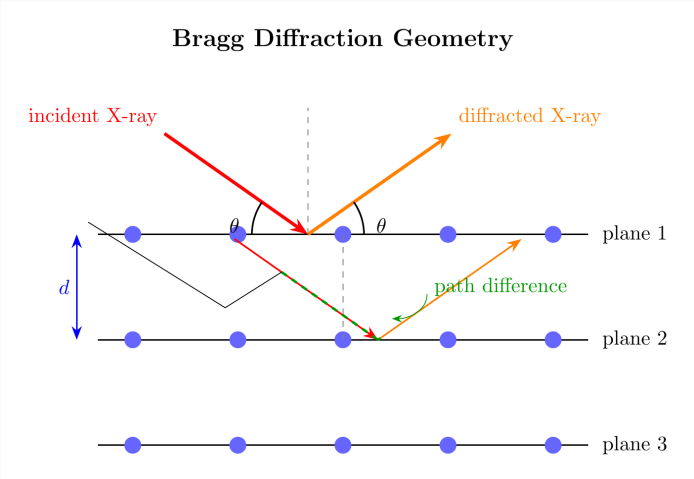

Fig. 1. A diagram illustrating Bragg's law geometry. Show an incident X-ray beam striking a set of parallel crystal planes at an angle θ. The planes are spaced by a distance d. Include the diffracted beam emerging at the same angle θ. Label the angle of incidence θ, the spacing between planes d, and indicate the path difference for constructive interference. The equation nλ = 2d sin θ should be included to show the condition for constructive interference. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly uses a specific numerical. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram labels the angle of incidence with. Regenerate using EXACTLY the values and sequences specified.

Question 2

10.0 points

Examine the diagram below of the reciprocal lattice and Ewald sphere construction. Determine where the reflections will occur when an X-ray beam is incident on a crystal. Define the conditions under which a specific point in the reciprocal lattice satisfies the Laue condition, and interpret its meaning in the context of X-ray diffraction.

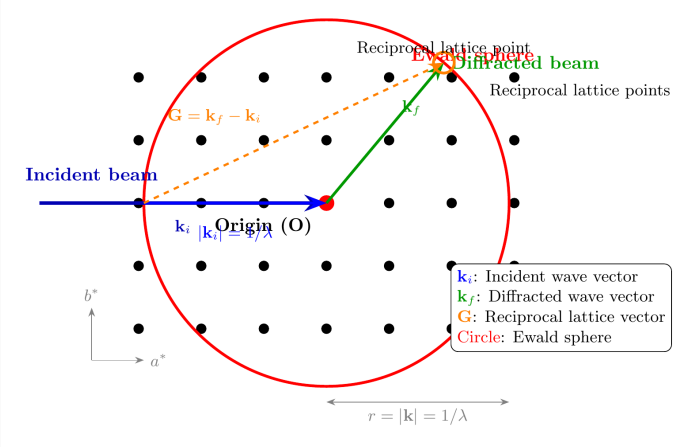

Fig. 2. A diagram showing the reciprocal lattice and Ewald sphere construction. The diagram should include: 1) a grid representing the reciprocal lattice points, 2) a circle representing the Ewald sphere, 3) an incident X-ray beam vector, 4) the origin of the reciprocal lattice at the center of the Ewald sphere, 5) a vector from the origin to a reciprocal lattice point indicating a possible reflection condition, and 6) labels for the incident beam, reciprocal lattice points, and Ewald sphere. IMPORTANT: The diagram is missing required labels. The diagram is missing a label. Ensure ALL structural components mentioned in the description are clearly labeled. IMPORTANT: The diagram is missing required labels. The diagram is missing a required label for the vector indicating. Ensure ALL structural components mentioned in the description are clearly labeled.

Question 3 10.0 points

Interpret the diagram below that illustrates Miller indices (100), (110), and (111) planes in a cubic unit cell. Explain how these planes are oriented with respect to the cubic axes and how each set of indices determines the plane's orientation.

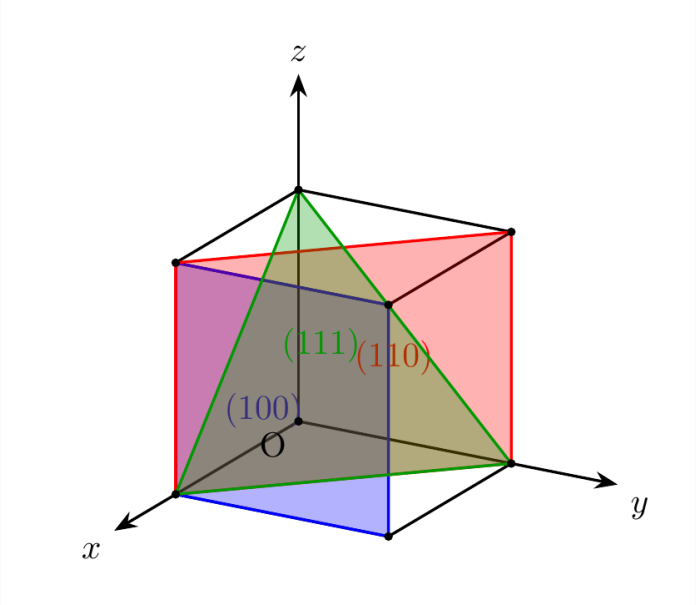

Fig. 3. A cubic unit cell showing the orientation of Miller indices planes (100), (110), and (111). The cubic unit cell should be drawn with its axes labeled as x, y, and z. The (100) plane should be shown as a plane perpendicular to the x-axis, the (110) plane should be shown as a plane cutting through the x and y axes, and the (111) plane should be shown as a plane cutting through the x, y, and z axes. Each plane should be distinctly labeled with its corresponding Miller indices. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly labels the (100. Regenerate using EXACTLY the values and sequences specified. IMPORTANT: Values or sequences in the diagram don't match the question. The diagram incorrectly represents the Miller indices planes. Regenerate using EXACTLY the values and sequences specified.

Question 4 10.0 points

Examine the diagram below of an FCC diffraction pattern. Identify the systematic absences in the diffraction peaks and explain the underlying cause of these absences based on the lattice plane reflections.

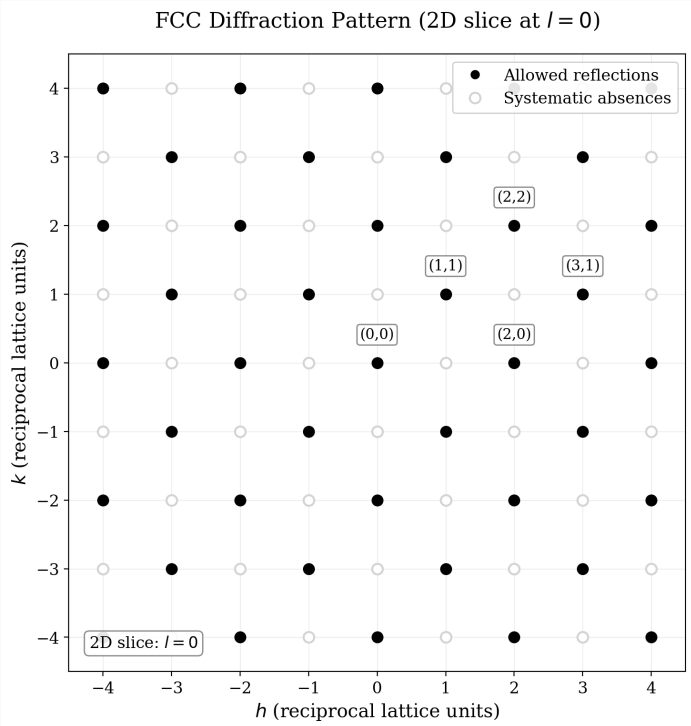

Fig. 4. CRITICAL: The previous diagram showed the wrong type. The diagram shows a 2D slice of a. Regenerate from scratch showing the CORRECT diagram type. Diagram of an FCC diffraction pattern showing diffraction peaks and highlighting systematic absences. The pattern should include labeled axes for reciprocal space, with peaks corresponding to allowed reflections and gaps where systematic absences occur due to lattice plane reflections. IMPORTANT correction needed: The diagram explicitly labels the "Systematic. Do NOT include region labels (Feasible/Infeasible, Stable/Unstable, Optimal, etc.), formula annotations, reaction product labels, threshold markers, or any text that reveals the answer. Show structure and context only.

Question 5 10.0 points

Examine the atomic arrangement of the BCC crystal in the diagram below. Use this to calculate the structure factor for a given set of Miller indices, assuming only atomic scattering factors are considered. Detail the steps involved in the calculation.

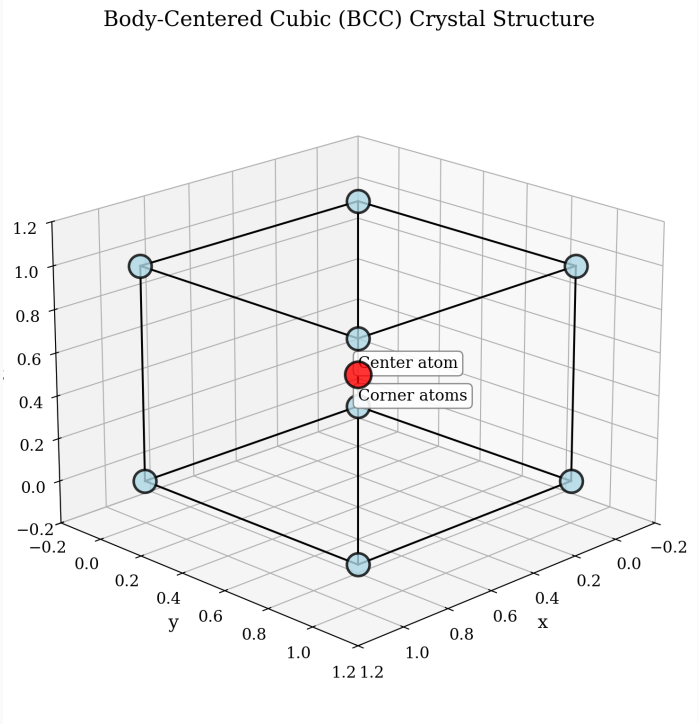

Fig. 5. 3D diagram of a BCC (body-centered cubic) unit cell showing the atomic arrangement. The diagram should include atoms at the corners of the cube and one atom at the center of the cube. The unit cell edges should be labeled, and the spatial arrangement should be clear to illustrate the BCC structure. This will help in understanding the calculation of the structure factor for given Miller indices.
IMPORTANT: The diagram is missing required labels. The diagram description requires the unit cell edges to be. Ensure ALL structural components mentioned in the description are clearly labeled.
IMPORTANT: The diagram is missing required labels. The diagram is missing the label for the z. Ensure ALL structural components mentioned in the description are clearly labeled.